

Renato Mascarenhas Pereira

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Experience

System Verification Engineer

May. 2025 - Present

SMITH+NEPHEW, COIMBRA, PORTUGAL

- Drive engineering activities and technical aspects of the program in partnership with the technical leads and program managers.
- Own DHF verification related documentation: Verification Plans, Verification Protocols, Test execution, and Verification Reports
- Support and review requirements development including System and sub-system Requirements.
- Define test methods for requirements specifications.
- Own JIRA defects repositories, log and prioritize software defects.
- Remain current with the systems engineering discipline and champion good systems engineering practice on development programs.
- Create and maintain new software and embedded test tools to enable automated testing.
- Own the test tool validation process and documentation, including the validation plans, and protocols.

Orthopedic Templates and Python Developer

Feb. 2022 - Jan. 2025

PEEKMED, BRAGA, PORTUGAL

- Prepared and processed 2D/3D CAD orthopedic templates (QCAD, megaCAD, MeshLab) and authored test cases, requirements, and detailed QA protocols for preoperative implant planning (verification & validation).
- Orchestrated full template lifecycle into development/staging/production, and validation, troubleshooting server-side errors and coordinating with the Servers team.
- Engineered and maintained Python scripts and Web apps to automate end-to-end orthopedic template workflows, providing ongoing technical support and reducing manual intervention.
- Managed version control with Bitbucket, tracked issues in JIRA, optimized database performance in MongoDB, and streamlined processes via VS Code and Terminal.

Customer Training

Sept. 2021 – Feb. 2022

ALERT LIFE SCIENCES COMPUTING, PORTO, PORTUGAL

- Scheduled and managed internal and customer training sessions, maintained training databases, and generated detailed reports.
- Designed and executed software test cases, performed functional QA of medical software, logged defects, and worked with developers to ensure quality.
- Prepared and delivered presentations to optimize learning outcomes.

Laboratory technician - Coordinator

May 2021 – Aug. 2022

PRO HEALTH MEDICAL, THE NETHERLANDS

- Coordinated laboratory operations, optimizing workflows, managing and scheduling multidisciplinary teams, overseeing inventory control, and ensuring timely instrument maintenance.
- Performed and validated both RT-qPCR and LAMP assays, conducted RNA/DNA extraction workflows, and executed comprehensive data analysis to deliver accurate, high-throughput COVID-19 test results.

Laboratory technician

May 2021 – Aug. 2022

UNILABS PORTUGAL, PORTO, PORTUGAL

- Performed and validated RT-qPCR assays, conducted RNA/DNA extraction workflows, and executed comprehensive data analysis to deliver accurate, high-throughput COVID-19 test results.

Education

Post-graduation in Clinical Trials

FACULTY OF MEDICINE, UNIVERSITY OF LISBON

2020 - 2021

Masters in Biomedical Engineering

FACULTY OF ENGINEERING OF THE UNIVERSITY OF PORTO

2018 - 2020

Bachelors in Bioengineering

FACULTY OF BIOTECHNOLOGY, CATHOLIC UNIVERSITY OF PORTUGAL

2015 - 2018

Research/Projects

i3S

RESEARCH TRAINEE / MASTER'S STUDENT FINALIST

Sept. 2017 – Oct. 2020

- Working in a research laboratory, BioEngineered Surfaces group, namely focusing on microbiological skills (BSL 2) and development of nanostructured surfaces for bacterial assays, host interactions and cancer.

CBQF

PROJECT LEADER / RESEARCH TRAINEE

June 2018 – Mar. 2020

- Responsible for algorithm/software development for detection of cardiovascular diseases using Machine Learning techniques and medical device development to final product.

Publications

- 2025 **Cholesterol Functionalized Nanoparticles Are Effective against *Helicobacter pylori*, the Gastric Bug: A Proof-of-Concept Study**,
R. Pereira, A. S. Pinho, M. Pereira, A. Rai, L. Ferreira, M. C. L. Martins, P. Parreira
- 2020 **Heart disease detection using ECG lead I and multiple pattern recognition classifiers**,
Pereira, R., Bispo, B., Rodrigues, P. M.

Continuous Education

- 2025 **OrthoMesh3D. Plan orthopaedic surgery in 3D using Blender**, Udemy 9.5h
- 2024 **Building and Deploying an AI-Powered Medical Imaging Chatbot**, Udemy 1.5h
- 2023 **Complete Python Developer in 2023 : Zero to Mastery**, Udemy 31h
- 2021 **3D for Medicine and Orthopedic Surgery**, Udemy 5h
- 2019 **Artificial Intelligence (AI) - Machine Learning for automatization and model prediction**, ESB-UCP 12h

Skills

Languages	Portuguese (native), English (proficient), Spanish (basic), Italian (beginner)
Management	Agile, JIRA, Project Planning, Team Leadership, Communication
Coding	Python (Pandas, Dash, VTK), Matlab, Arduino, Git, Medical Image
3D Modeling & Printing	SolidWorks, Shapr3D, InVesalius, OsiriX, Rhino, MeshMixer, UltiMaker Cura
Laboratory Techniques	Microbiological & Cellular cultures, Characterization, Analytical Chemistry, Medical instrumentation

Honor and Awards

- 2021 **Merit Award, Scholarship**, Direção-Geral do Ensino Superior, DGES
- 2015 **Merit Award, Scholarship**, Novo Banco